Text Steganography with Dynamic Codebook and Multimodal Large Language Model

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Introduction

Generative Text Steganography

Generative Text Steganography (GTS) hides secret bits during language-model generation, producing fluent and natural-looking stego text. Its key challenge is achieving a good trade-off among **capacity, quality, and imperceptibility**.

Limitations of Existing Methods

White-box methods (e.g., ADG and Discop)

White-box methods need shared models and access to token probabilities.

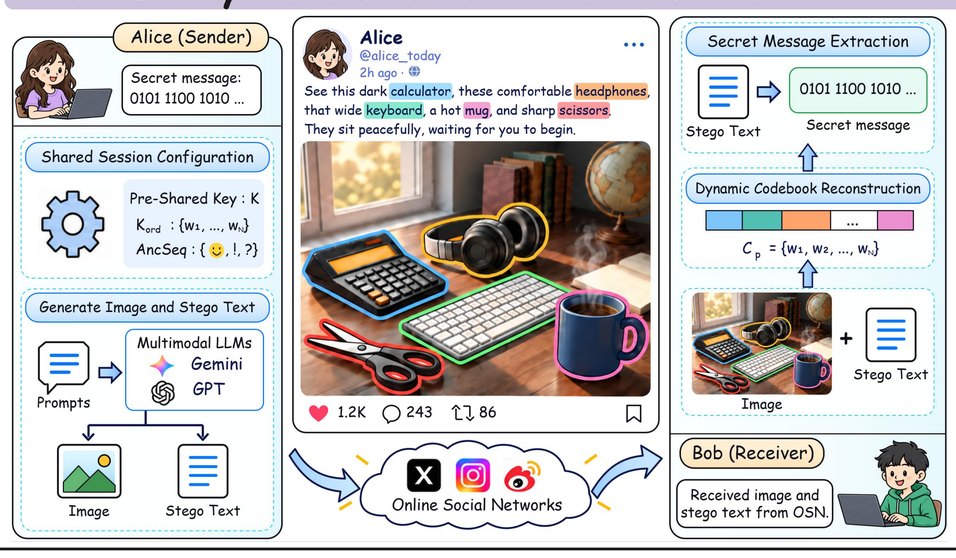
Black-box methods (e.g., LLM-Stega)

Prior black-box methods rely on reusable codebooks and per-sentence prompts.

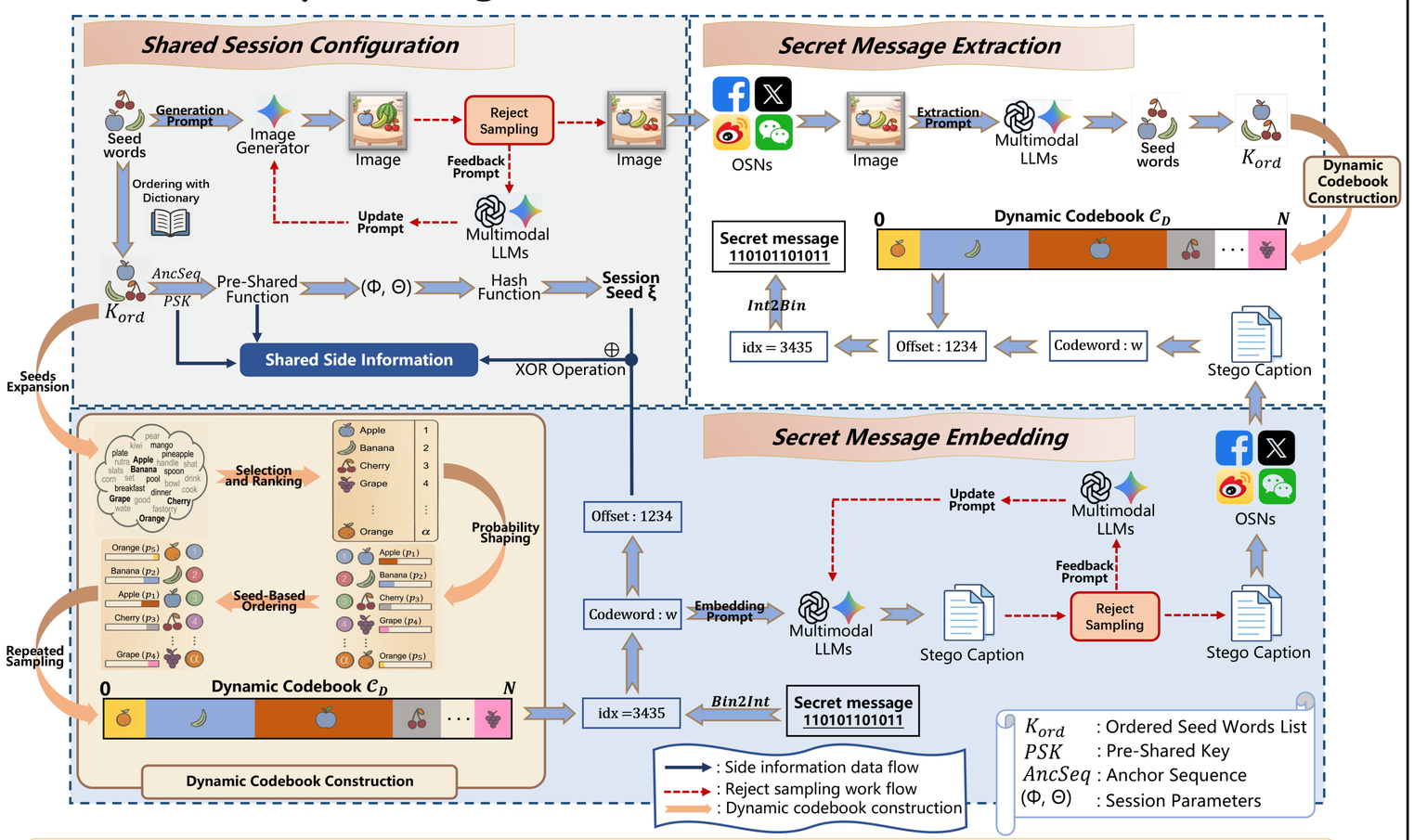
Our Approach: Dynamic Codebook

Public image-caption context and a pre-shared key are used to **construct a dynamic codebook for each session**, improving flexibility in black-box MLLM settings while preserving image-text consistency.

Case Study on Online Social Networks



Method: DyCo-Stega



The overall framework of DyCo-Stega. Crucially, the dynamic codebook construction is executed independently by both parties without transmitting the codebook itself.

1. Shared Session Configuration

Alice uses the ordered seed words to guide a multimodal LLM in generating a semantic image. Through reject sampling, the image is iteratively refined until the same seed words can be reliably recovered by the MLLM. Together with the pre-shared key and anchor sequence, these seed words define the session parameters and provide a stable public visual anchor that Bob can later reproduce.

2. Dynamic Codebook Construction

The recovered seed words are expanded using a public dictionary to form a session-specific semantic pool. Candidate words are selected and ranked by dictionary frequency, assigned probabilities through a half-Gaussian shaping function, and then permuted by a PRNG initialized with the session seed ξ. Finally, repeated sampling maps the reordered words to contiguous intervals over [0, N), producing the dynamic codebook.

3. Secret Message Embedding

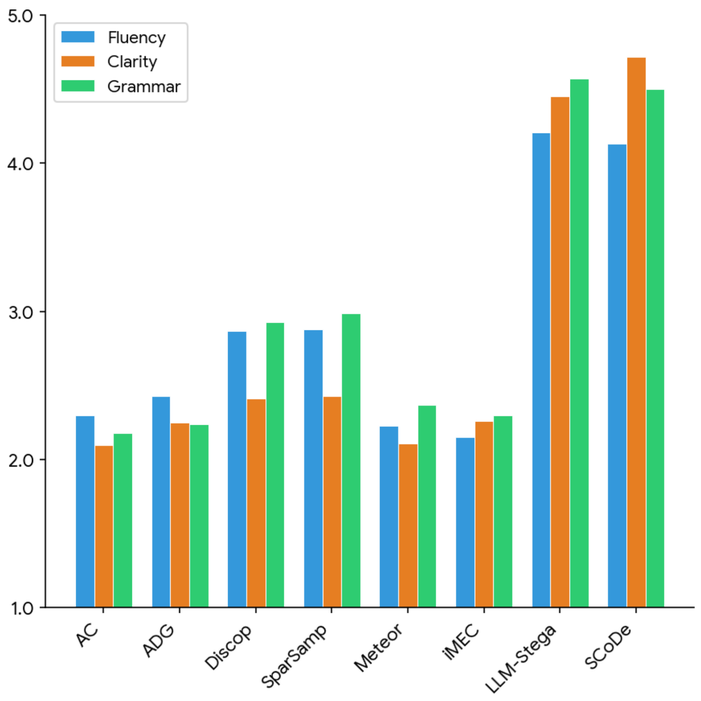
The secret bits are first converted into an integer index within [0, N). The codebook interval containing this index determines the corresponding codeword, while the distance from the interval boundary defines the offset. The offset is XOR-masked with the session seed ξ and transmitted through the side channel. The MLLM then generates an image-aligned stego caption containing the codeword and anchor sequence, with reject sampling used to enforce the embedding constraints.

4. Secret Message Extraction

Bob retrieves the public image and stego caption from the social network, recovers the seed words with the MLLM, and recomputes the same session seed ξ. Using the recovered seed words, anchor sequence, and pre-shared key, Bob independently reconstructs the identical dynamic codebook. He then identifies the codeword interval, restores the offset by undoing the XOR mask, and combines them to recover the original secret-message index and bit sequence.

Experiments

Human evaluation



Main results

Method	PPL↓	SS↑	KLD↓	EC↑	Acc
White-box (GPT-2)					
AC	410.0	0.6980	164.20	3.89	0.5805
ADG	876.9	0.7070	45.11	5.33	0.4893
Discop	396.4	0.5500	11.42	3.80	0.4908
SparSamp	377.9	0.5540	10.87	3.91	**0.5000**
METEOR	298.3	0.5490	30.83	2.42	0.4910
iMEC	589.1	0.5740	76.71	3.07	0.4928
White-box (Qwen2.5)					
AC	380.0	0.4360	51.43	4.16	0.7698
ADG	461.3	0.6480	29.87	4.86	0.5403
Discop	145.3	0.6910	10.45	3.63	0.5595
SparSamp	140.0	0.6950	10.41	3.77	0.5150
METEOR	243.1	0.6800	24.57	3.72	0.5188
iMEC	405.9	0.6560	36.22	3.46	0.5340
Black-box (GPT-5.1)					
LLM-Stega	137.8	0.7230	5.81	5.27	0.5073
DyCo-Stega	**128.9**	**0.8940**	**3.37**	**6.37**	0.4997

Image-text consistency

Scene	CLIP (Cover)	CLIP (Stego)	Δ
Office Desk	32.99	33.02	0.03
Kitchen Counter	30.00	30.16	0.16
Bathroom Vanity	32.23	32.22	−0.01
Average	**31.74**	**31.80**	**+0.06**

Extension to Video

Video Generator	Video Recognizer				
	Gemini 3.1 Flash	Gemini 3.1 Pro	GPT 5.3	GPT 5.4(Std)	GPT 5.4(Ext)
Veo	89%	**95%**	73%	82%	84%
Sora	82%	84%	70%	77%	78%
Seedance	83%	87%	68%	76%	80%

Video Generator	Video Recognizer				
	Gemini 3.1 Flash	Gemini 3.1 Pro	GPT 5.3	GPT 5.4(Std)	GPT 5.4(Ext)
Veo	139	173	**133**	207	212
Sora	205	261	197	294	298
Seedance	541	583	516	619	633

Ablation and robustness studies

(a) Cross-Model Robustness

Gen.	Rec.	S@1	S@3	AvgR
Gemini	Gemini	83%	98%	1.21
	GPT	80%	97%	1.23
	Grok	74%	96%	1.28
	Claude	81%	97%	1.25
GPT	Gemini	76%	94%	1.31
	GPT	72%	93%	1.36
	Grok	74%	92%	1.33
	Claude	78%	93%	1.31
Qwen	Gemini	60%	68%	1.24
	GPT	55%	65%	1.27
	Grok	52%	64%	1.28
	Claude	59%	66%	1.25

(b) Probability Shaping Ablation

Setting	PPL ↓	KLD ↓	SS ↑
Half-Gaussian ($\sigma=\tau/4$)	105.7	30.32	0.903
Half-Gaussian ($\sigma=\tau/2$) ★	128.9	**3.37**	0.894
Half-Gaussian ($\sigma=\tau$)	167.9	13.40	0.836
Half-Gaussian ($\sigma=2\tau$)	179.1	19.73	0.814
Half-Gaussian ($\sigma=4\tau$)	182.4	38.21	0.854
Uniform	247.0	12.35	0.805

(c) OSN Robustness

Platform	S@1	S@2	S@3
WeChat	97%	100%	100%
Instagram	98%	99%	100%
Facebook	98%	100%	100%
X	97%	99%	100%
WeiBo	98%	99%	100%

(d) Prompt & Temperature

Temperature	Full	Mini
0	100%	40.7%
0.7	100%	48.7%
1.0	99.3%	39.3%
1.5	94.0%	34.7%
2.0	90.8%	30.2%

Conclusion

DyCo-Stega introduces a black-box text steganography framework that constructs dynamic codebooks from public visual context, eliminating reliance on fixed codebooks.

Experiments demonstrate strong performance in text quality, embedding capacity, imperceptibility, anti-steganalysis, and secret recovery, with robust performance across different models and online social network transmission.

The framework also shows potential for cross-modal extension, including video and future applications to audio.

Read Paper

Code

Acknowledgments

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